PRATHAP SELVAKUMAR

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PERSONAL STATEMENT

MSc Robotics student at the University of Manchester with a strong foundation in AI, computer vision, and autonomous systems. Experienced in designing and deploying intelligent robotics pipelines using Python, ROS2, and deep learning frameworks. Published researcher with industry experience across software engineering and AI/ML.

WORK EXPERIENCE

C-Square Info Solutions (Reliance Retail Subsidiary) Aug 2024 – May 2025
Software Engineer · Remote

- Re-engineered **50+ static APIs** into dynamic, modular endpoints, eliminating over **1,000 redundant lines of code** per report page and improving system **scalability by 40%**.
- Redesigned front-end architecture using reusable **React hooks** and **context APIs**, increasing interface responsiveness and maintainability by **25%**.
- Optimised database queries and implemented secure **RESTful API authentication**, reducing data retrieval time by **30%**.

Qentelli Solutions Feb 2024 – Apr 2024
AI/ML Intern · Hyderabad

- Designed and deployed **TensorFlow-based automation models**, reducing manual task effort by **40%**.
- Processed and optimised over **100,000 training data entries**, improving pipeline speed and data quality by **35%**.
- Evaluated multiple classification models, achieving **92% accuracy** on predictive tasks

National University of Singapore Dec 2023 – Jan 2024
Global Academic Intern — Big Data Analytics using Deep Learning · Singapore

- Led a **team of 4 interns** to deliver a deep learning project, achieving a final score of **44/50 (Grade A)**.
- Trained and fine-tuned **neural networks** on large-scale datasets, improving model precision by **15%**.
- Built Python-based data visualisation dashboards (**Matplotlib, Seaborn**), reducing analysis time by **25%**.

PROJECTS

Autonomous Rover Development (Leo Rover) Oct 2025 – May 2026

- Designed and deployed a **ROS2-based robotic system** achieving **95% autonomous navigation accuracy** in indoor and semi-outdoor environments.
- Collaborated in a **5-member team** to calibrate control algorithms, reducing **positional drift by 20%**.
- Built a **RealSense D435i computer vision pipeline** for coloured block detection using **HSV masking**, computing **3D drop coordinates** for robotic arm manipulation.

Drone Feedback Controller — Mar 2026 – Apr 2026

- Designed a hybrid **Q-learning** and **nested PID control system** for **UAV stabilisation**, with an online **RL agent** selecting optimal gain profiles in real time from discretised flight states.
- Implemented experience replay using a **400-sample circular buffer** for stable **off-policy TD (0)** updates, with adaptive integral clamping tuned for **wind disturbance rejection**.

A* Path Planning System — Mar 2026 – Apr 2026

- Implemented an interactive **A* path planning system** with dynamic obstacle placement, route replay, and **heuristic-based shortest-path search**.
- Built a **React/TypeScript** interface with **live grid resizing**, path validation, and cost diagnostics.

Agro Analytics — Agricultural AI Pipeline — Dec 2023 – Jan 2024

- Built an **AI-driven crop monitoring pipeline** using **satellite, soil, and sensor data** for predictive agricultural analysis.
- Developed **Random Forest models** that improved crop yield forecasting accuracy by **12%** over baseline methods.

Snake Detection — Computer Vision — Jul 2022 – Aug 2022

- Trained a custom **YOLOv8** model for **real-time snake detection** and venomous/non-venomous classification with **94% confidence**.
- Integrated **GPS-based alert logging** to support rapid location-aware safety notifications.

Audio Search Engine — Published Research — Apr 2022 – May 2022

- Developed an audio search engine using **unsupervised clustering** and **feature extraction** for similarity-based retrieval across large datasets.
- Published **peer-reviewed research** in **IGI Global** on scalable audio indexing: "Comparative Analysis Implementation of Queuing Songs in Players Using Audio Clustering Algorithm".

TECHNICAL SKILLS

- **Robotics & Simulation:** ROS2, PyBullet, Gazebo, MoveIt2, Nav2, Intel RealSense D435i
- **AI / Machine Learning:** TensorFlow, PyTorch, Stable-Baselines3, YOLOv8, Scikit-learn, OpenCV
- **Programming:** Python, C++, MATLAB, JavaScript (React, Node.js), TypeScript
- **Cloud & Data:** AWS SageMaker, Pandas, Matplotlib, Seaborn, Excel (advanced, pivot tables)
- **Other:** Git, RESTful APIs, Agile/Scrum, sensor integration, algorithm design, image processing
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EDUCATION

University of Manchester — 2025 – 2026 (Expected)

MSc Robotics

- Relevant modules: **Control Systems**, **Multi-Agent Reinforcement Learning**, **Autonomous Systems**, Digital Manufacturing, **Computer Vision**.
- Used **MATLAB**, **TensorFlow**, and **OpenCV** to simulate and automate robotic tasks, reducing testing time by **30%**.

SRM Institute of Science and Technology — 2020 – 2024

BEng Computer Science and Engineering — First Class with Distinction, CGPA: 8.49 / 10

- Applied **AI, machine learning, and data science** methodologies across **six applied research** and development projects.
- Built an **AI-based image recognition system** for automation, increasing classification accuracy by **12%** and reducing manual verification time by **25%**.

PUBLICATIONS

Selvakumar, P. (2022). *Comparative Analysis Implementation of Queuing Songs in Players Using Audio Clustering Algorithm.* IGI Global. Peer-reviewed journal article.